

SUPPLEMENTARY MATERIAL

Ghost Infrastructure: Historical Industrial Geography and the Persistence of Path Dependent Accessibility in Bochum, Germany

Figures S1-S6 — Essen replication and threshold-sensitivity supplementary figures

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These figures accompany the manuscript's main-text figures 1-6 and are referenced inline at the corresponding points in the manuscript text.

Figure S1. Study Area — Essen, North Rhine-Westphalia, Germany



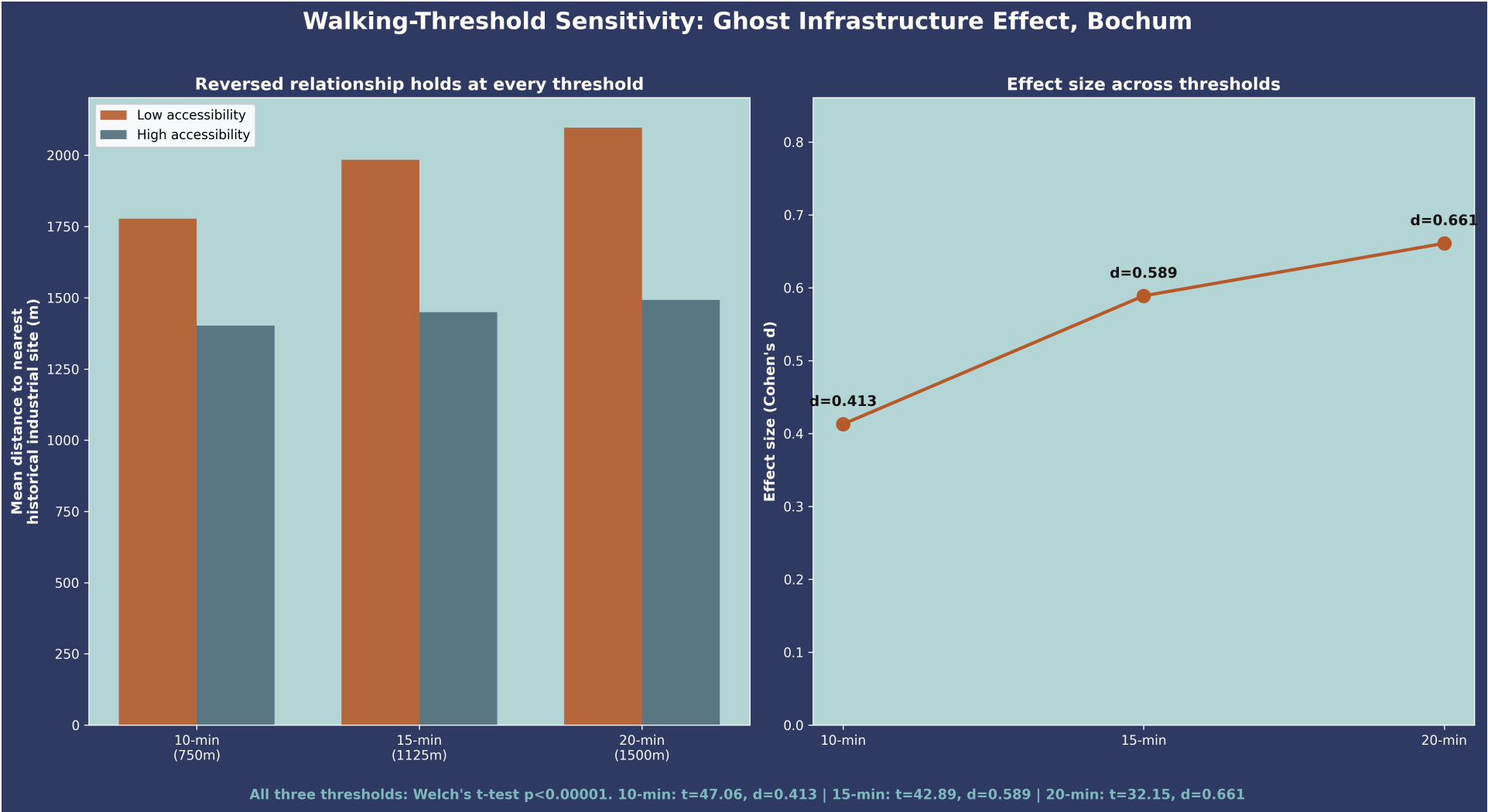
Map of Essen's administrative boundary in North Rhine-Westphalia, Germany, directly comparable to Figure 1 (Bochum)

Figure S2. Essen's Historical Coal-Mining Geography, 1779-1986



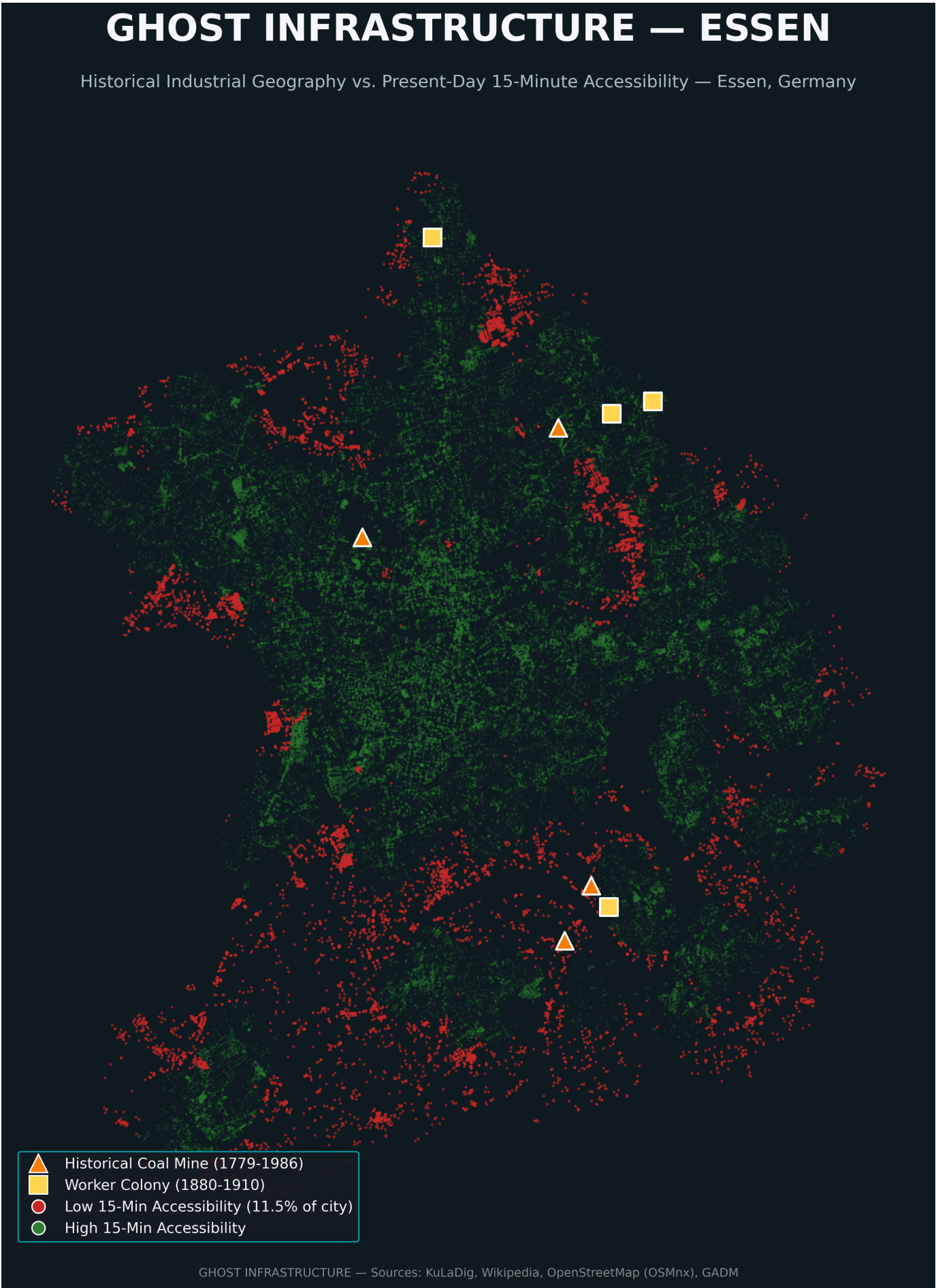
Essen's historical coal mining geography: 4 coal mines and 4 worker colonies, digitized from KuLaDig and German Wikipedia, directly comparable to Figure 2 (Bochum).

Figure S3. Walking-Threshold Sensitivity: Ghost Infrastructure Effect, Bochum



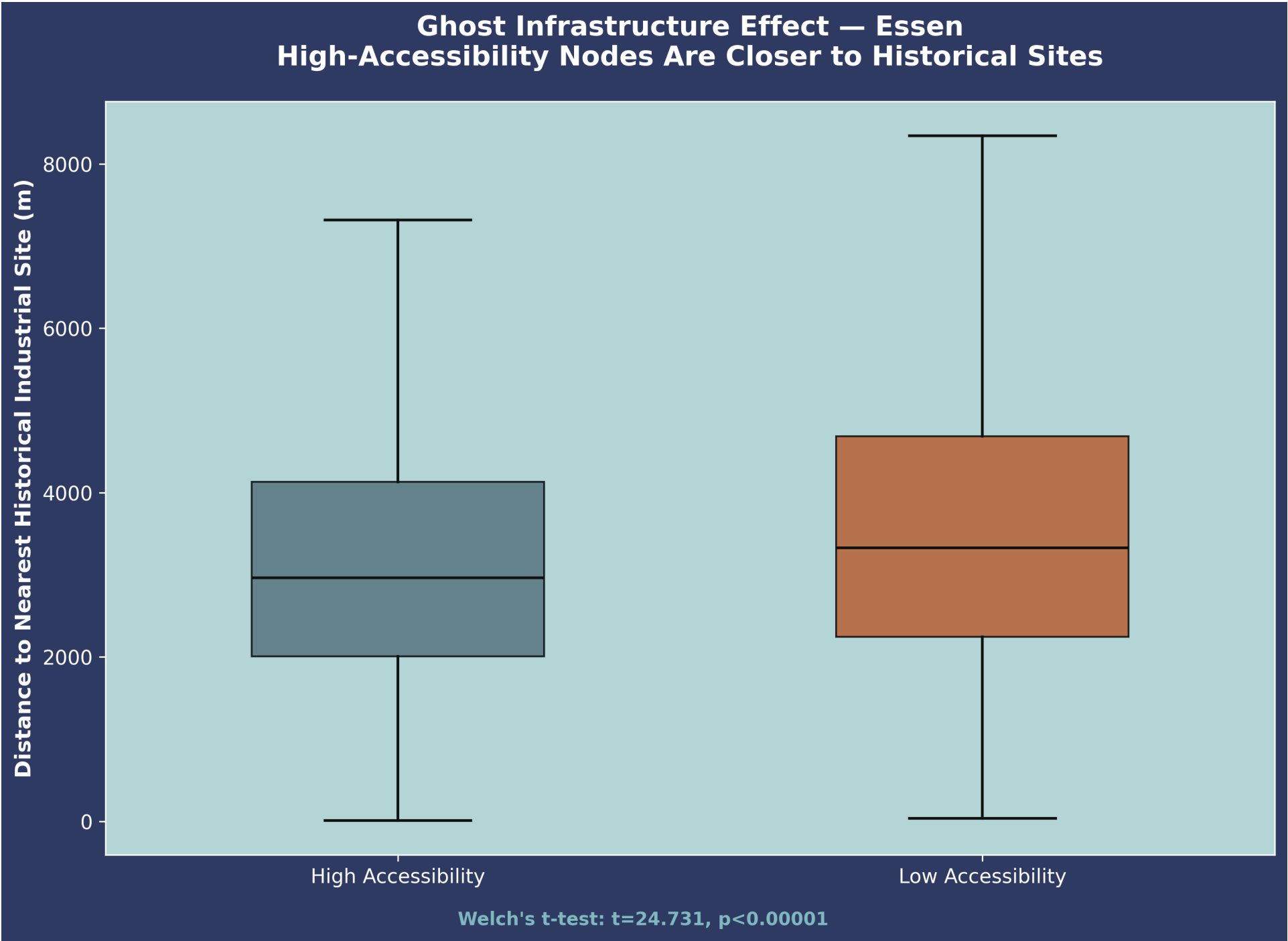
Effect size comparison of distance to nearest historical site by accessibility group (left) and mean distance to nearest historical site at 10, 15, and 20 minute thresholds (right).

Figure S4. Historical Geography vs. Present-Day 15-Minute Accessibility — Essen



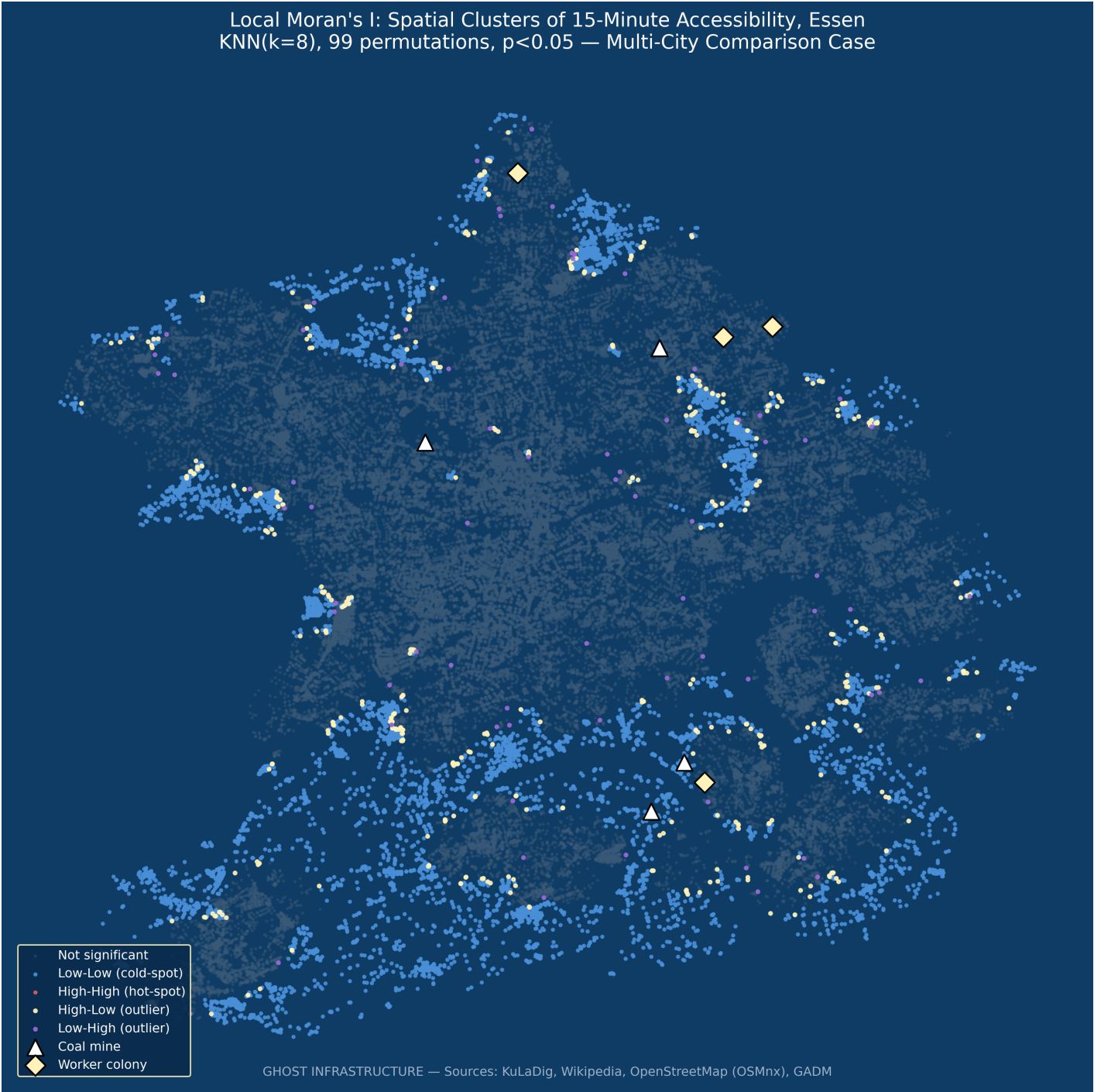
Historical industrial sites overlaid on present day 15 minute walking accessibility in Essen, directly comparable to Figure 3 (Bochum)

Figure S5. Distance to Nearest Historical Site: High vs. Low Accessibility — Essen



Distance to historical sites for high and low accessibility nodes in Essen, comparable to Figure 4 (Bochum).

Figure S6. Local Moran's I Cluster Map — Essen



Local Moran's I cluster map for Essen, directly comparable to Figure 5 (Bochum).